

ORION

PART 01

ORION Core Vision

Overall purpose, mission, scope, principles

Technical Architecture Series • Beginner → Professional

How to use this volume

This document is one independent volume in the twelve-part ORION technical series. Beginners should read from the start; experienced readers can jump directly to the relevant architecture sections.

ORION is a secure communications, infrastructure-coordination, emergency-response, and decision-support architecture. High-impact weapons-control functionality is deliberately outside this design.

What ORION is

ORION is a proposed resilient digital infrastructure platform for secure communication, emergency coordination, infrastructure telemetry, and AI-assisted decision support.

It is a federation of cooperating systems, not one unrestricted central computer.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Why it exists

Modern networks are fragmented. A disaster can simultaneously disrupt connectivity, power, transport, cloud services, and emergency coordination.

ORION creates a common coordination layer and alternative communication paths.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Mission

Preserve essential communication during disruption.

Give authorized operators a shared operational picture.

Automate bounded, low-risk coordination while retaining human authority over high-impact decisions.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Core principles

Resilience before convenience.

Least privilege.

Explicit trust boundaries.

Graceful degradation.

Observable and auditable actions.

Replaceable components and open interfaces.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

System vision

Citizens and responders → applications → identity and policy → communication and event fabric → AI analysis → terrestrial/NTN/edge networks → external infrastructure.

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Scope boundary

In scope: communication resilience, emergency workflows, identity, policy, telemetry, AI decision support, simulation, observability.

Out of scope: autonomous weapon operation, targeting, launch authority, or other direct harmful-force control.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Success metrics

Availability, recovery time, emergency message delivery, unauthorized-action rate, AI accuracy/calibration, detection time, operator workload, and cost per active user.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Architecture Review Checklist

- Requirements and assumptions are explicit.
- Component ownership and boundaries are documented.
- Normal, degraded, and failure states are defined.
- Security and privacy are architectural requirements.
- Capacity and scaling assumptions are measurable.
- Logs, metrics, traces, and audit events cover critical flows.
- Dependencies have explicit contracts and failure behavior.
- Important decisions record alternatives and trade-offs.
- Testing covers expected behavior and controlled failures.
- Operational runbooks exist for important incidents.

Key Takeaways

ORION Core Vision should be implemented as a bounded layer of ORION, with explicit interfaces to the rest of the platform.

The goal is not maximum complexity. The goal is measurable resilience, security, maintainability, interoperability, and operational usefulness.

Glossary

- API — Application Programming Interface used for software-to-software communication.
- Edge — Computing placed near users or physical systems.
- Event — A record that something happened.
- Failover — Switching to a redundant component after failure.
- Idempotency — Repeating an operation does not create an unintended additional effect.
- NTN — Non-terrestrial network, commonly involving satellite infrastructure.
- Policy — A rule that determines what is allowed under stated conditions.
- RPO — Recovery Point Objective; acceptable data-loss window.
- RTO — Recovery Time Objective; target recovery time.
- Zero trust — Continuous verification instead of implicit trust based on network location.